Start Here

when you are stuck

THE PROCESS
1. **Read the question again.** Write down what it asks for.
2. **Name the type.** Is this a linear equation, a factoring problem, a system?
3. **Pick the method** that matches that type.
4. **Work one step per line.** Never do two moves in one line.
5. **Check the answer.** Put it back into the original problem.

Pre-Algebra

fractions

THE PROCESS
1. **Factor** the top and the bottom.
2. **Cancel** every factor they share.
3. **Multiply** what remains.

THE SAME SUM, BOTH WAYS

$\frac{1}{4}+\frac{1}{6}$ by the general rule: $\frac{(1)(6)+(1)(4)}{(4)(6)}=\frac{10}{24}=\frac{5}{12}$

by the smallest common denominator, 12: $\frac{3}{12}+\frac{2}{12}=\frac{5}{12}$

Same answer. The general rule needs no thinking and may need simplifying. The smallest denominator needs thinking and usually does not.

ratios and proportions

THE PROCESS — SOLVE A PROPORTION
1. **Write** the proportion, keeping matching units in matching positions.
2. **Cross multiply** — top left times bottom right, and the other pair.
3. **Solve** the resulting simple equation.

exponents and square roots

THE RULE
1. **Move the point** until exactly one non-zero digit sits in front of it.
2. **Count the moves.** That count is the exponent.
3. **Moved left** gives a positive exponent. **Moved right** gives a negative one.

Algebra I

solving linear equations

THE PROCESS
1. **Clear brackets** with the distributive rule.
2. **Combine like terms** on each side.
3. **Move all variable terms** to one side, using the balance rule.
4. **Move all plain numbers** to the other side.
5. **Divide** by the number multiplying the variable.
6. **Check** by substituting into the original equation.

turning words into algebra

THE PROCESS
1. **Read it once** without writing anything.
2. **Name the unknown.** Write "let $x=$" and then the thing in words.
3. **Translate one phrase at a time** into symbols.
4. **Solve** the equation.
5. **Answer the question asked**, with units.

inequalities

THE FLIP RULE

If you **multiply or divide both sides by a negative number**, reverse the inequality sign.

$$-2x>8 \quad \Longrightarrow \quad x<-4$$

Adding or subtracting a negative does not flip it. Only multiplying or dividing does.

THE TEST
1. **Pick** any easy number inside your answer, usually 0.
2. **Put it into the original** inequality.
3. **It must be true.** Then pick a number outside, which must be false.

absolute value

THE TWO-CASE RULE

$|A|=b$ with $b>0$ splits into two equations:

$$A=b \quad \text{or} \quad A=-b$$

Isolate the absolute value first. Split only when the bars stand alone on one side.

systems of equations

Which method? — look at the equations

A **substitution** — one equation already has a variable alone ($y=\ldots$), or is easy to make so

B **elimination** — matching terms line up, so adding or subtracting cancels one variable

WORKED (A): $y=2x-1$ **AND** $3x+y=9$

y is already alone, so replace y in the second equation:

$3x+(2x-1)=9 \;\Rightarrow\; 5x-1=9 \;\Rightarrow\; 5x=10 \;\Rightarrow\; x=2$

put $x=2$ back into $y=2x-1$: $\quad y=4-1=3$

solution $(2,\,3)$

↓

WORKED (B): $2x+3y=12$ **AND** $2x-y=4$

both have $2x$, so subtract the second from the first and the x terms cancel:

$$(2x-2x)+(3y-(-y))=12-4 \;\Rightarrow\; 4y=8 \;\Rightarrow\; y=2$$

put $y=2$ into $2x-y=4$: $\quad 2x-2=4 \;\Rightarrow\; x=3$

solution $(3,\,2)$

CHECK BOTH EQUATIONS, NOT ONE

$2(3)+3(2)=6+6=12$ ✓ $\qquad 2(3)-2=4$ ✓

A point can satisfy one equation and fail the other. Checking only one proves nothing.

one solution	the lines cross once
no solution	parallel lines. You reach a false statement such as $0=5$.
infinitely many	the same line twice. You reach $0=0$.

THE PROCESS

1. **Pick** one variable to remove.
2. **Eliminate** it from one pair of equations.
3. **Eliminate** the same variable from a different pair.
4. **Solve** the two-variable system you now have.
5. **Back-substitute** to find the third variable.

factoring

Which method? — count the terms

terms	shape	method
2	a^2-b^2	difference of squares
2	$a^3\pm b^3$	sum or difference of cubes
3	x^2+bx+c	find two numbers
3	$ax^2+bx+c,\ a\neq 1$	the AC method
4	anything	grouping

difference of squares	$a^2-b^2=(a+b)(a-b)$
perfect square	$a^2\pm 2ab+b^2=(a\pm b)^2$
sum of cubes	$a^3+b^3=(a+b)(a^2-ab+b^2)$
difference of cubes	$a^3-b^3=(a-b)(a^2+ab+b^2)$

A sum of squares a^2+b^2 does not factor. Only the difference does.

WORKED (2 TERMS): $4x^2-25$

both parts are perfect squares: $4x^2=(2x)^2$ and $25=5^2$

$$4x^2-25=(2x+5)(2x-5)$$

↓

WORKED (3 TERMS, $a=1$**):** x^2+5x+6

find two numbers that multiply to 6 and add to 5

pairs for 6: 1 and 6 give 7. $\quad$ 2 and 3 give 5 ✓

$$x^2+5x+6=(x+2)(x+3)$$

↓

WORKED (3 TERMS, $a\neq 1$**):** $6x^2+11x+3$ **— THE AC METHOD**

1. multiply a and c: $\quad 6\cdot 3=18$
2. find two numbers that multiply to 18 and add to 11: $\quad 9$ and 2
3. split the middle term using them: $\quad 6x^2+9x+2x+3$
4. group in pairs and factor each: $\quad 3x(2x+3)+1(2x+3)$
5. the bracket now repeats, so pull it out:

$$(3x+1)(2x+3)$$

Check any factoring by multiplying back out. It takes ten seconds and catches every sign error.

quadratic equations

Which method? — look at the equation

A **factoring** — the quadratic factors easily. Fastest when it works.

B **square root property** — there is no plain x term, or one side is already a perfect square

C **completing the square** — asked for it, or you need **vertex form**

D **quadratic formula** — always works. Use it when nothing factors.

Try A first, fall back to D. B and C are for the shapes that ask for them.

THE ZERO PRODUCT RULE — WHY FACTORING SOLVES

If two numbers multiply to 0, at least one of them is 0. Nothing else can happen.

So once $(x-2)(x-3)=0$, either $x-2=0$ or $x-3=0$.

This works for zero only. If $(x-2)(x-3)=6$, you may not say $x-2=6$. Move everything to one side first.

WORKED (A) FACTORING: $x^2-5x+6=0$

two numbers multiplying to 6, adding to -5: $\quad -2$ and -3

$(x-2)(x-3)=0$, so $x-2=0$ or $x-3=0$

$$x=2 \quad \text{or} \quad x=3$$

↓

WORKED (B) SQUARE ROOT PROPERTY: $(x-3)^2=16$

take the square root of both sides, and keep both signs:

$$x-3=\pm 4$$

$x-3=4$ gives $x=7$. $\quad x-3=-4$ gives $x=-1$.

$$x=7 \quad \text{or} \quad x=-1$$

Forgetting the negative root loses half the answer. A square erases the sign, so undoing it must restore both.

↓

WORKED (C) COMPLETING THE SQUARE: $x^2+6x-7=0$

1. move the plain number across: $\quad x^2+6x=7$
2. halve the x coefficient and square it: $\quad \left(\frac{6}{2}\right)^2=3^2=9$
3. add it to both sides: $\quad x^2+6x+9=16$
4. the left side is now a perfect square: $\quad (x+3)^2=16$
5. square root both sides: $\quad x+3=\pm 4$, so $x=1$ or $x=-7$

↓

WORKED (D) THE FORMULA: $2x^2+3x-5=0$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \qquad a=2,\ b=3,\ c=-5$$

1. the part under the root: $\quad b^2-4ac=3^2-4(2)(-5)=9+40=49$
2. substitute: $\quad x=\frac{-3\pm\sqrt{49}}{2(2)}=\frac{-3\pm 7}{4}$
3. split the $\pm$: $\quad x=\frac{4}{4}=1 \quad$ or $\quad x=\frac{-10}{4}=-\frac{5}{2}$

$-4ac$ with a negative c becomes addition. Two negatives made $+40$ here.

Algebra II

rational expressions

THE PROCESS — SIMPLIFY
1. **Factor** the top and the bottom completely.
2. **Note the excluded values** — whatever makes the original bottom zero.
3. **Cancel** factors the top and bottom share.

exponential and logarithmic functions

Which method? — solving these two shapes

A **variable in the exponent** ⇒ take a logarithm of both sides

B **variable inside a log** ⇒ combine into one log, then rewrite as an exponential

WORKED (A), THE EASY CASE: $2^x=32$

write 32 as a power of the same base: $32=2^5$

same base means the exponents match: $x=5$

(Always look for a common base first. It avoids the calculator entirely.)

↓

WORKED (A), THE GENERAL CASE: $3^x=20$

20 is not a power of 3, so take a log of both sides:

$$\log(3^x)=\log 20 \;\Rightarrow\; x\log 3=\log 20 \;\Rightarrow\; x=\frac{\log 20}{\log 3}\approx 2.727$$

The power rule is what frees the x: it moves the exponent down in front.

↓

WORKED (B): $\log_2 x+\log_2(x-2)=3$

1. combine with the product rule: $\log_2\big(x(x-2)\big)=3$

2. rewrite as an exponential: $x(x-2)=2^3=8$

3. solve: $x^2-2x-8=0 \;\Rightarrow\; (x-4)(x+2)=0 \;\Rightarrow\; x=4$ or $x=-2$

4. check both. $x=-2$ gives $\log_2(-2)$, and a logarithm of a negative number does not exist.

$x=4$ only

Log equations almost always produce a rejected answer. Checking is part of the method, not an optional extra.

sigma notation and the binomial theorem

THE PROCESS — EXPAND $(a+b)^n$
1. **Take** row n of the triangle as the coefficients.
2. **Start** at $a^n b^0$ and step down: the power of a falls, the power of b rises.
3. **Check** that every term's powers add to n.

direct and inverse variation

THE PROCESS — THE SAME THREE STEPS EITHER WAY
1. **Write** the correct form, $y=kx$ or $y=\frac{k}{x}$.
2. **Find** k by substituting the pair you were given.
3. **Rewrite** with k filled in, then answer the question.

polynomial division

THE REMAINDER THEOREM

To find the remainder when $P(x)$ is divided by $(x-c)$, just evaluate $P(c)$. No division needed.

Consequence: if $P(c)=0$, the remainder is zero, so $(x-c)$ is a factor. This is how you find roots of a cubic.

finding polynomial roots

THE RATIONAL ROOT THEOREM

Every rational root of a polynomial has the form

$$\pm\frac{\text{a factor of the constant term}}{\text{a factor of the leading coefficient}}$$

That list is finite, so you can test it. It lists candidates, not answers — most of them fail.

THE PROCESS
1. **List** the candidates from the theorem.
2. **Test** them with the remainder theorem until one gives 0.
3. **Divide** the polynomial by that factor.
4. **Factor** what is left, which is now one degree lower.

Pre-Calculus

functions

THE TEST — SUBSTITUTE $-x$ AND COMPARE

$f(x)=x^2-4$: $f(-x)=(-x)^2-4=x^2-4$. That equals $f(x)$, so even.

$f(x)=x^3-x$: $f(-x)=-x^3+x=-(x^3-x)$. That equals $-f(x)$, so odd.

Compare against both $f(x)$ and $-f(x)$ before concluding "neither".

THE PROCESS
1. **Replace** $f(x)$ with y.
2. **Swap** x and y.
3. **Solve** for y.
4. **Rename** it $f^{-1}(x)$.

polynomial and rational behaviour

Asymptotes — three kinds

An **asymptote** is a line the graph approaches but never reaches.

V **vertical** ⇒ where the denominator is zero after cancelling. A factor that cancels gives a **hole**, not an asymptote.

H **horizontal** ⇒ compare degrees. Top smaller: $y=0$. Equal: the ratio of leading coefficients. Top larger: none.

S **slant** ⇒ when the top degree is exactly one more than the bottom. Divide to find the line.

WORKED: $\frac{2x^2+3x-1}{x^2-4}$

vertical: $x^2-4=0$ gives $x=2$ and $x=-2$, and neither cancels

horizontal: degrees are equal, both 2, so take the leading coefficients:

$$y=\frac{2}{1}=2$$

slant: none, because the degrees are equal rather than differing by one

A graph may cross a horizontal asymptote. The rule is about behaviour far out, not a barrier.

finding the zeros

THE PROCESS
1. **Set the function equal to zero.**
2. **Solve** by whichever method fits the type.
3. **Discard** any answer outside the domain.
4. **Check** each survivor in the original function.

Which method? — read the function type

A **linear** ⇒ solve directly. Exactly one zero, unless the line is horizontal.

B **quadratic** ⇒ factor, or use the formula. The discriminant says how many before you start.

C **polynomial, degree** ≥ 3 ⇒ rational root theorem, then divide and factor what is left.

D **rational** ⇒ set the numerator to zero, then discard anything that also zeroes the denominator.

E **exponential** ⇒ no zeros ever. b^x is always positive, which is why $y=0$ is its asymptote.

F **logarithm** ⇒ set the inside equal to 1, because $\log 1=0$.

G **trigonometric** ⇒ infinitely many, spaced one period apart. Give the ones in the interval asked for.

ONE EXAMPLE OF EACH

A $3x-12=0 \;\Rightarrow\; x=4$

B $x^2-6x+5=0 \;\Rightarrow\; (x-1)(x-5)=0 \;\Rightarrow\; x=1,\,5$

C $x^3-4x=0 \;\Rightarrow\; x(x-2)(x+2)=0 \;\Rightarrow\; x=0,\,2,\,-2$

F $\log(x-3)=0 \;\Rightarrow\; x-3=1 \;\Rightarrow\; x=4$

G $\sin x=0$ on $[0,2\pi) \;\Rightarrow\; x=0,\,\pi$

CASE D IS THE TRAP — $f(x)=\dfrac{x^2-4}{x-2}$

numerator zero at $x=2$ and $x=-2$. But $x=2$ also makes the denominator zero.

$x=2$ is a **hole**, not a zero — the function has no value there at all.

$$\text{the only zero is } x=-2$$

Always check a candidate against the denominator before accepting it.

Always factor first. A factored function shows its zeros directly: each factor set to zero gives one, by the zero product rule.

the key points of a graph

THE PROCESS
1. **y-intercept** — set $x=0$. There is at most one, because a function gives one output per input.
2. **x-intercepts** — set $y=0$ and find the zeros. There may be many.
3. **Turning points** — where the curve changes direction.
4. **Asymptotes** — lines the curve approaches but never reaches.
5. **End behaviour** — what happens far left and far right.

sketching a polynomial

THE PROCESS
1. **Factor** and list the zeros. Those are the only places the curve can cross.
2. **Check each zero's multiplicity** — does the curve cross there, or touch and turn back?
3. **Find the y-intercept** by setting $x=0$.
4. **Read the end behaviour** from the leading term.
5. **Test one point in each interval** between the zeros to get the sign.
6. **Join the dots.** A turning point sits between each pair of neighbouring zeros.

polynomial and rational inequalities

THE PROCESS
1. **Move everything to one side**, so the other side is zero.
2. **Factor** and find the zeros. For a fraction, also find where the denominator is zero.
3. **Mark all of those** on a number line. They split it into intervals.
4. **Test one point** in each interval.
5. **Keep the intervals** whose sign matches what was asked.

trigonometric identities and equations

THE ONE IDENTITY EVERYTHING ELSE COMES FROM

$$\sin^2\theta+\cos^2\theta=1$$

This is the Pythagorean theorem on the unit circle: the legs are $x=\cos\theta$ and $y=\sin\theta$, and the hypotenuse is the radius 1.

THE PROCESS — SOLVE A TRIG EQUATION
1. **Isolate** the trig function.
2. **Find** one angle that works, using a special triangle.
3. **Find every other angle** in the interval with the same value.
4. **Check** each against the interval you were given.

triangles that are not right-angled

Which law? — count what you are given

A **Law of Sines** ⇒ you have a matching angle and opposite side. Cases **AAS**, **ASA**, **SSA**.

B **Law of Cosines** ⇒ you do not. Cases **SAS** and **SSS**.

The test is one question: can you see an angle and the side directly across from it, both known? If yes, use Sines. If no, use Cosines.

Law of Sines $\qquad \dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}$

Law of Cosines $\qquad c^2=a^2+b^2-2ab\cos C$

Each side pairs with the angle opposite it. Lower-case a sits across from upper-case A.

The Law of Cosines is the Pythagorean theorem with a correction term. When $C=90°$, $\cos C=0$, the last term vanishes, and it becomes $c^2=a^2+b^2$ exactly.

WORKED (A): $A=30°$, $B=45°$, $a=10$

a and A are a matching pair, so use Sines:

$$\frac{10}{\sin 30°}=\frac{b}{\sin 45°} \implies b=\frac{10\sin 45°}{\sin 30°}=\frac{10\left(\frac{\sqrt{2}}{2}\right)}{\frac{1}{2}}$$

$$b=10\sqrt{2}\approx 14.14$$

Sanity check: B is the larger angle, so b must be the longer side. $14.14>10$ ✓

↓

WORKED (B): $a=5$, $b=7$, $C=60°$

The known angle sits between the two known sides, so no matching pair exists. Use Cosines:

$$c^2=5^2+7^2-2(5)(7)\cos 60°=25+49-70\left(\tfrac{1}{2}\right)=74-35=39$$

$$c=\sqrt{39}\approx 6.24$$

Sanity check: c must sit between $7-5=2$ and $7+5=12$. It does ✓

SSA can give two triangles, one, or none. Two different angles share the same sine, so after finding one, check whether its supplement also fits inside $180°$. This is called the **ambiguous case**.